ASSESSMENT 5

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QUESTION:

A diagnostic laboratory is investigating differences in cell-nucleus characteristics between benign and malignant samples. Using the **Breast Cancer Wisconsin dataset** available in scikit-learn, examine mean radius, mean texture, mean smoothness, and mean concavity.

Select a reproducible, class-balanced random sample of **200 observations (100 benign and 100 malignant)** using a fixed random seed of 42. Apply appropriate feature scaling and develop two complementary multivariate visualizations to investigate differences between the diagnostic categories. Interpret the observed patterns, similarities, and overlapping characteristics.

[10 Marks]

SOLUTION:

[1]:
```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

[2]:
```
data = load_breast_cancer(as_frame=True)
df = data.data
```

[3]:
```
features = [
    'mean radius',
    'mean texture',
    'mean smoothness',
    'mean concavity'
]

X = df[features].copy()
y = data.target
```

[4]:
```
benign = X[y == 1].sample(n=100, random_state=42)
malignant = X[y == 0].sample(n=100, random_state=42)

sample = pd.concat([benign, malignant])
labels = pd.Series(
    ['Benign'] * 100 + ['Malignant'] * 100,
    index=sample.index
)
```

[5]:
```
sample = sample.sample(frac=1, random_state=42)
labels = labels.loc[sample.index]

print("Sample size:", len(sample))
print("\nClass distribution:")
print(labels.value_counts())
```

[6]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(sample)

scaled_df = pd.DataFrame(
    X_scaled,
    columns=features,
    index=sample.index
)
```

[7]:

```
plt.figure(figsize=(11, 6))

positions = [1, 2, 4, 5, 7, 8, 10, 11]
box_data = []

for feature in features:
    box_data.append(
        scaled_df.loc[labels == 'Benign', feature]
    )
    box_data.append(
        scaled_df.loc[labels == 'Malignant', feature]
    )

plt.boxplot(box_data, positions=positions, widths=0.7)

plt.xticks(
    [1.5, 4.5, 7.5, 10.5],
    features,
    rotation=20
)

plt.ylabel('Standardized Value (z-score)')
plt.title('Feature Distributions by Diagnostic Category')
plt.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.show()
```

[8]:
```
pca = PCA(n_components=2)
principal_components = pca.fit_transform(X_scaled)

plt.figure(figsize=(9, 6))

for category, color in [('Benign', 'blue'),
                        ('Malignant', 'red')]:
    mask = (labels.values == category)

    plt.scatter(
        principal_components[mask, 0],
        principal_components[mask, 1],
        label=category,
        alpha=0.7
    )

plt.xlabel(
    f'PC1 ({pca.explained_variance_ratio_[0] * 100:.1f}% variance)'
)
plt.ylabel(
    f'PC2 ({pca.explained_variance_ratio_[1] * 100:.1f}% variance)'
)
plt.title('PCA: Benign vs Malignant Samples')
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

[9]:
```
comparison = scaled_df.groupby(labels).mean().round(2)

print("\nMean standardized feature values:")
print(comparison)

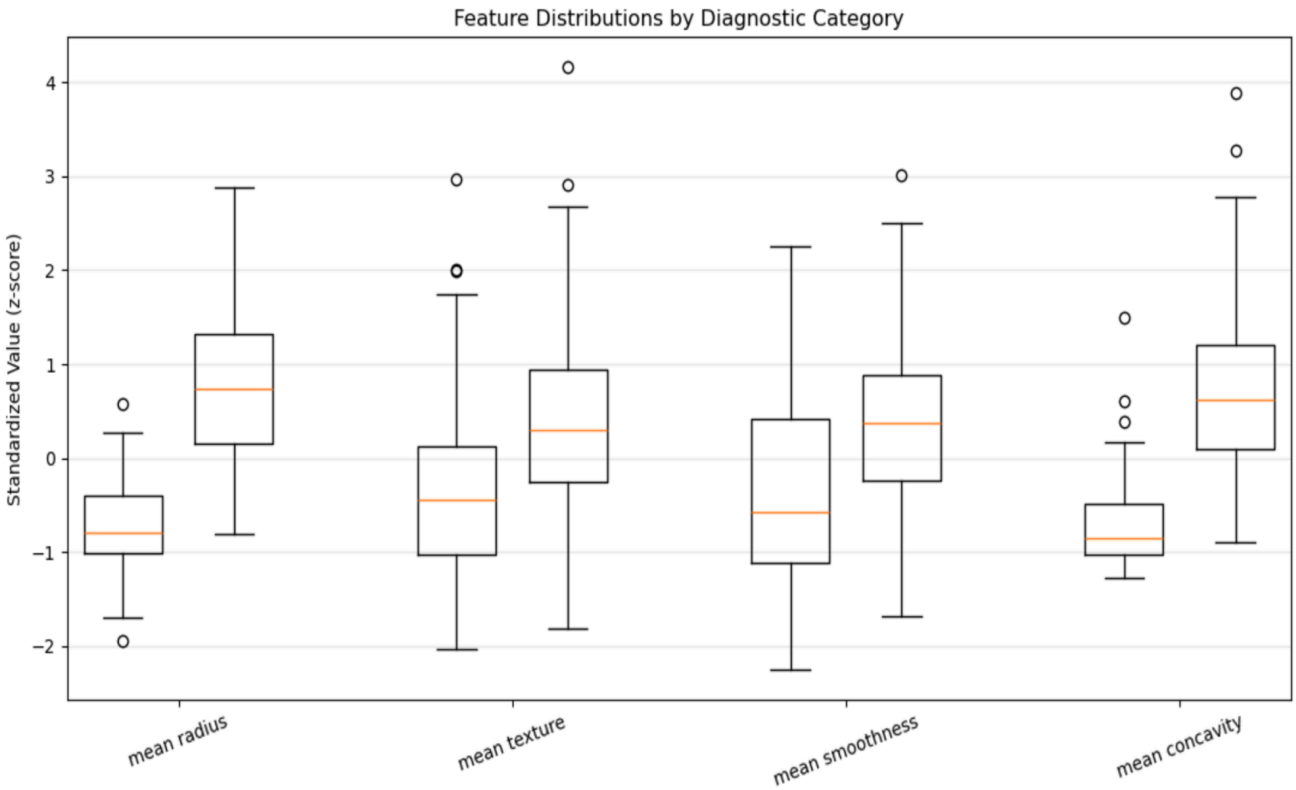
print("\nTotal variance explained by two PCs:",
      round(pca.explained_variance_ratio_.sum() * 100, 2), "%")
```

OUTPUT:

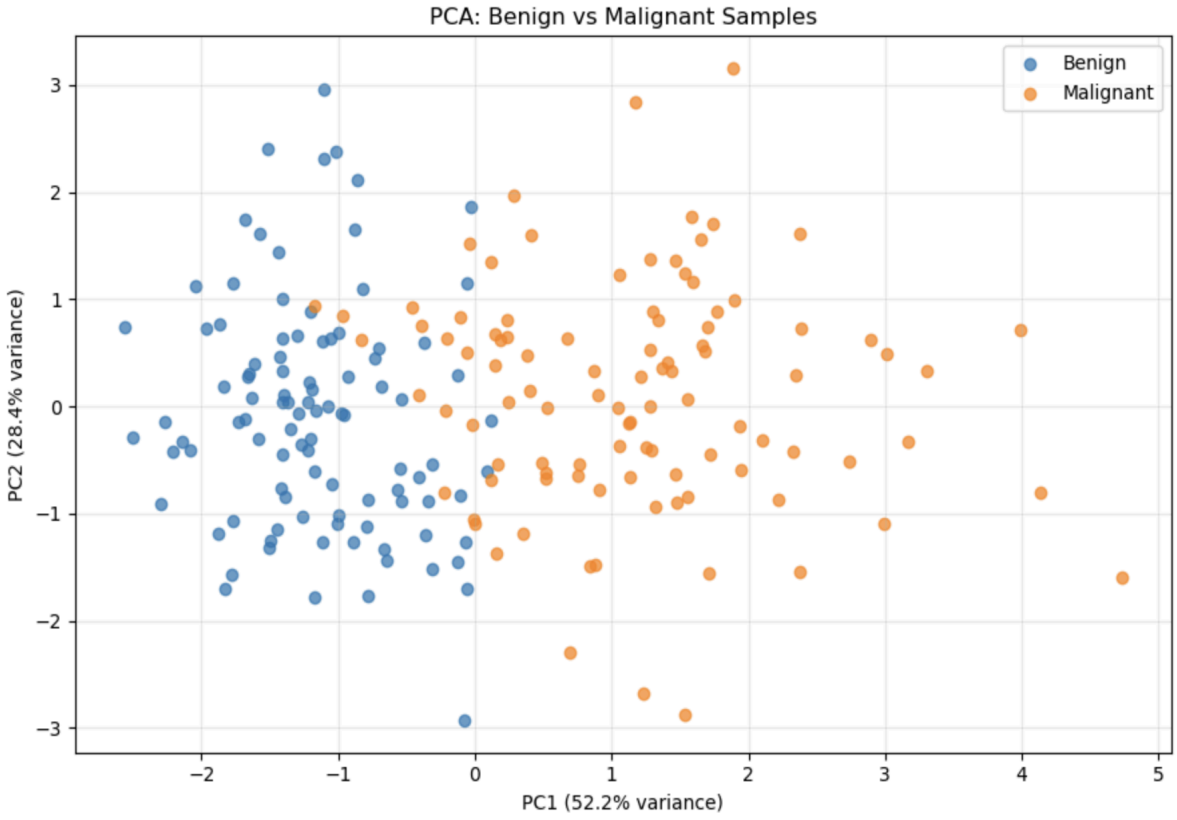
```
Sample size: 200

Class distribution:
Benign       100
Malignant    100
Name: count, dtype: int64
```

Standardised BoxPlots:



PCA scatter plot:



Comparison:

Mean standardized feature values:

	mean radius	mean texture	mean smoothness	mean concavity
Benign	−0.74	−0.34	−0.39	−0.72
Malignant	0.74	0.34	0.39	0.72

Total variance explained by two PCs: 80.61 %

INTERPRETATION:

Standardized Boxplots: Malignant samples generally show higher mean radius and mean concavity than benign samples. Some overlap exists in all four features, indicating individual variation and limited ability of a single feature to distinguish the categories.

PCA Scatter Plot: The plot shows differences between benign and malignant samples based on their combined features. Overlapping points indicate similar characteristics between some samples, while separated clusters suggest differences between the diagnostic categories.

Conclusion: The visualisations reveal differences and similarities in cell-nucleus characteristics between benign and malignant samples.